

Apache *Kaylin*

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Set Up Kaylin

First Come Up!!!

01

Quick Guide Installation Apache Kaylin

This one about how to get Kaylin from the website

02

Docker Container Set Up

After we download and foldering the Kaylin we need to Run the Docker Container (Contain Tips and Trick if Their is A Problem)

03

Set Up OLAP

How we to set up in Apache Kaylin di web page

Quick Guide Installation

Images Contain:



Hadoop 2.7.0



Jdk 1.8



Hive 1.2.1



Hbase 1.1.2



Spark 2.3.1



Zookeeper 3.4.6



Kafka 1.1.1



Mysql

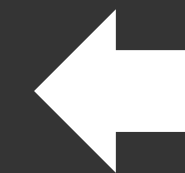


Maven 3.6.1

How We Started

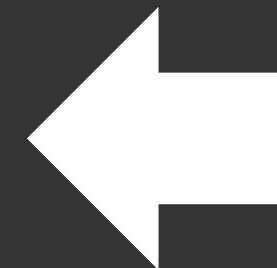
Open Windows Power Shell

```
docker pull apachekylin/apache-kylin-standalone:3.0.1
```



This command is use to
pull/download file folder of
Apache Kylin

```
docker run -d \  
-m 8G \  
-p 7070:7070 \  
-p 8088:8088 \  
-p 50070:50070 \  
-p 8032:8032 \  
-p 8042:8042 \  
-p 16010:16010 \  
apachekylin/apache-kylin-standalone:3.0.1
```



After Download Complete Execute
This One

Please make sure you have a good connection

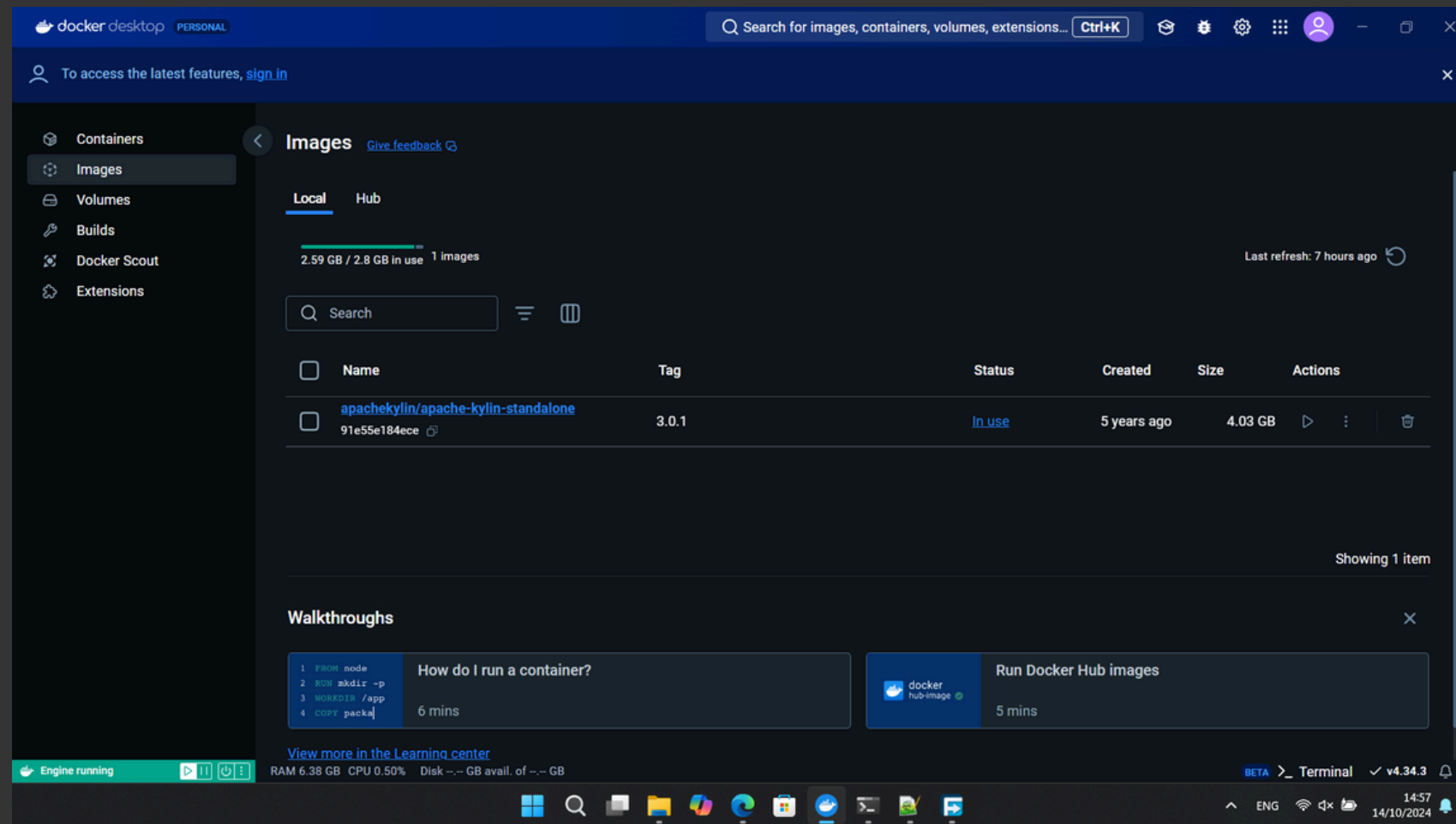
```
Windows PowerShell x Windows PowerShell x + v
3.0.1: Pulling from apachekylin/apache-kylin-standalone
eef58c7f98e9: Download complete
4a00b6a3dc44: Download complete
de24ae1dc934: Download complete
4da2bcf28ce2: Download complete
d43b7a000177: Download complete
155f1ec45c79: Download complete
638c7082ba84: Download complete
f2a74e1fcf1a: Download complete
fb7f8f2c76f7: Download complete
ef7cd122ff67: Download complete
4218deff225a: Download complete
e16a6df3203f: Download complete
43ab431abd59: Download complete
d3d888485fe6: Download complete
8a205d5ca91b: Download complete
60620bc83e11: Download complete
cedf84f50916: Download complete
d7ce67937c31: Download complete
cbac6436a996: Download complete
640e40760c9f: Download complete
0f0ff3699671: Download complete
0fa067284b01: Download complete
Digest: sha256:91e55e184ece32324a08020443e8c062b4429061905de1800273b2984b842456
Status: Downloaded newer image for apachekylin/apache-kylin-standalone:3.0.1
docker.io/apachekylin/apache-kylin-standalone:3.0.1

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview apachekylin/apache-kylin-standalone:3.0.1
PS C:\Users\zulfi> docker run -d -m 8G -p 7070:7070 -p 8088:8088 -p 50070:50070 -p 8032:8032 -p 8042:8042 -p 16010:16010 apach
e kylin/apache-kylin-standalone:3.0.1
```

Preview After Pulling Apache Kylin

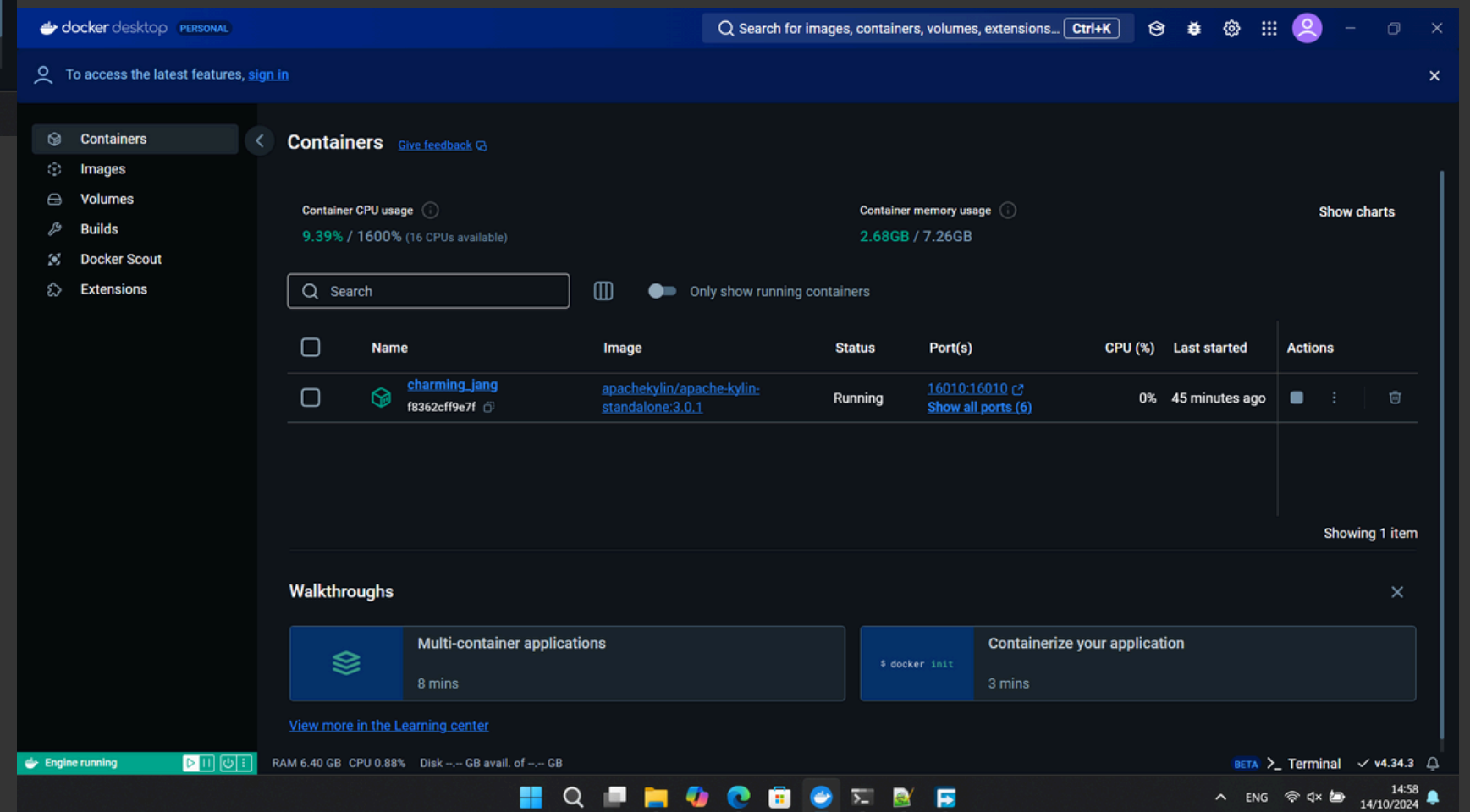
Docker Set Up

After you finished pull the Apache Kaylin, IT will automaticaly display and run in your container and image



There is 2 Problem that common happen.

- 1.Exit 139
- 2.Port are used



EXIT 139

When Exit 139 happens the docker image cannot run

Step 1 : Create a file named `.wslconfig` in your user directory

Ex: `C:\Users\zulfi`

Step 2 : Add following line to the `.wslconfig` file

`[wsl2]`

`kernelCommandLine = vsyscall=emulate`

Step 3 : Restart Your PC

Please edit or create the file you can use Text Editor such as Notepad++

Source From : <https://stackoverflow.com/questions/67126007/apache-kylin-docker-image-fails-to-start>

Port Used

Step 1 : Open Terminal or Window Powershell (As an Administrator)

Step 2: Turn of and Turn on Windows Network Address Translation

Execute this ->

```
net stop winnat
```

```
net start winnat
```

Source From : <https://stackoverflow.com/questions/65272764/ports-are-not-available-listen-tcp-0-0-0-0-50070-bind-an-attempt-was-made-to>

Set Up Kaylin

First one you need to open the container which contain Apache Kaylin by click the name of the container

Start Kaylin

The screenshot shows the Docker Desktop interface. The top bar includes the Docker logo, 'docker desktop PERSONAL', a search bar, and various icons. Below the top bar, there's a navigation sidebar with icons for Containers, Images, Volumes, Extensions, and Settings. The main area displays the 'Containers' tab, showing system usage (CPU: 10.62%, Memory: 4.43GB) and a table of running containers. One container, 'charming_jang', is highlighted with a red box. To the right, a 'Notifications' panel shows an error message: '(HTTP code 500) server error - Ports are not available: exposing port TCP 0.0.0.0:50070 -> 0.0.0.0:0: listen tcp 0.0.0.0:50070: bind: An attempt was made to access a socket in a way forbidden by its access permissions.' The bottom status bar shows system resources (RAM, CPU, Disk) and the Docker version (v4.34.3).

docker desktop PERSONAL

Search for images, containers, volumes, extensions... Ctrl+K

To access the latest features, [sign in](#)

Containers [Give feedback](#)

Container CPU usage 10.62% / 1600% (16 CPUs available)

Container memory usage 4.43GB / 7.26GB

Show charts

Search

Only show running containers

	Name	Image	Status	Port(s)	CPU (%)	Last started	Actions
☐	charming_jang f8362cff9e7f	apachekylin/apache-kylin-standalone:3.0.1	Running	16010:16010 Show all ports (6)	10.39%	1 hour ago	

Showing 1 item

Walkthroughs

Multi-container applications
8 mins

`$ docker init` Containerize your application
3 mins

[View more in the Learning center](#)

RAM 5.67 GB CPU 1.13% Disk -- GB avail. of -- GB

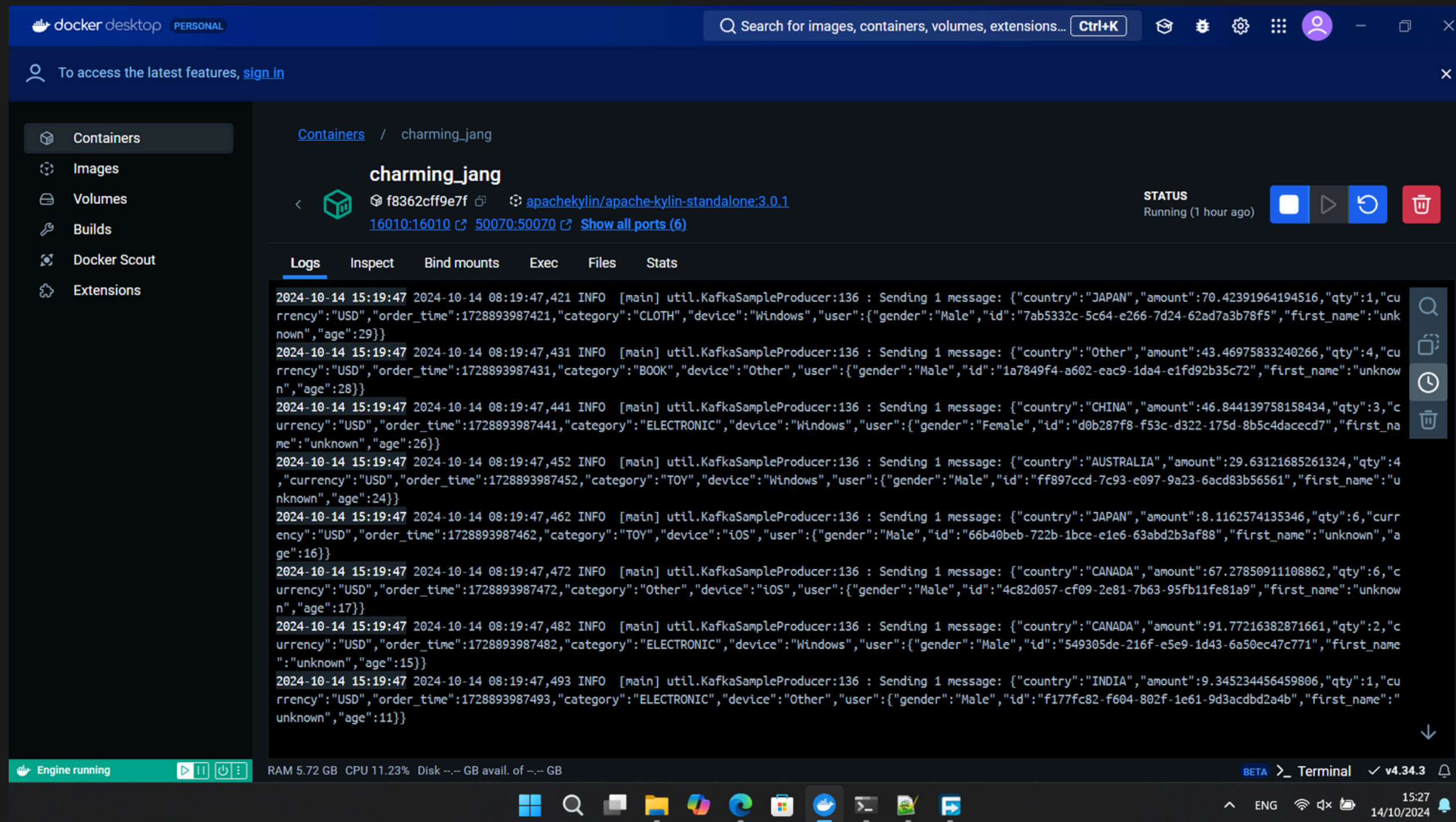
BETA > Terminal ✓ v4.34.3

ENG 15:23 14/10/2024

(HTTP code 500) server error - Ports are not available: exposing port TCP 0.0.0.0:50070 -> 0.0.0.0:0: listen tcp 0.0.0.0:50070: bind: An attempt was made to access a socket in a way forbidden by its access permissions.
4 hours ago

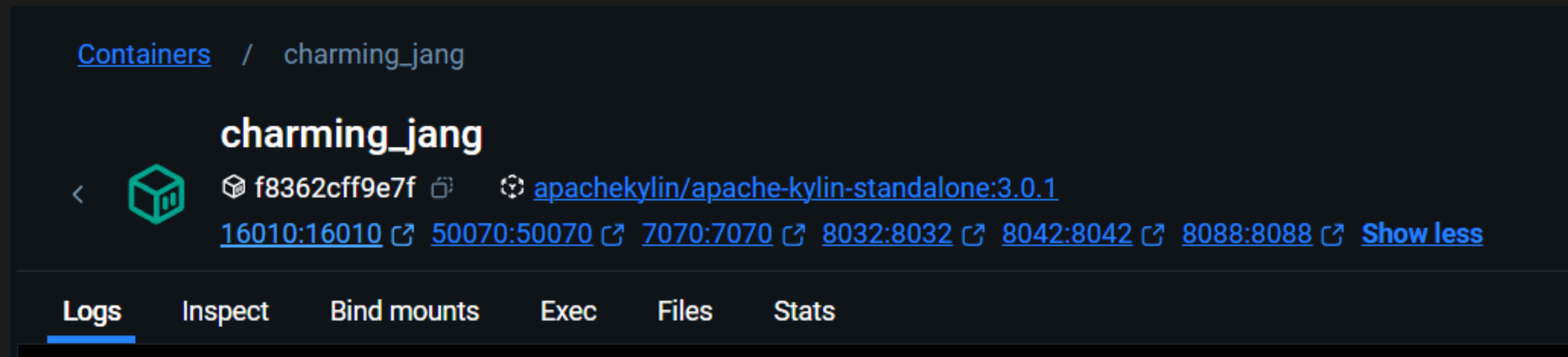
First thing that you can see is the container logs

- Container logs will display what happen in your contain for example the stream data from Kafka



In the upper section you will see some port which will navigate you to every part contain in image such as:

1. Kylin Page: <http://127.0.0.1:7070/kylin/>
2. HDFS NameNode Page: <http://127.0.0.1:50070>
3. Yarn ResourceManager Page: <http://127.0.0.1:8088>
4. HBase Page: <http://localhost:16010/master-status>
5. Hadoop - Node Information : <http://localhost:8042/>
6. Hadoop - All Application Information : <http://localhost:8088/cluster>

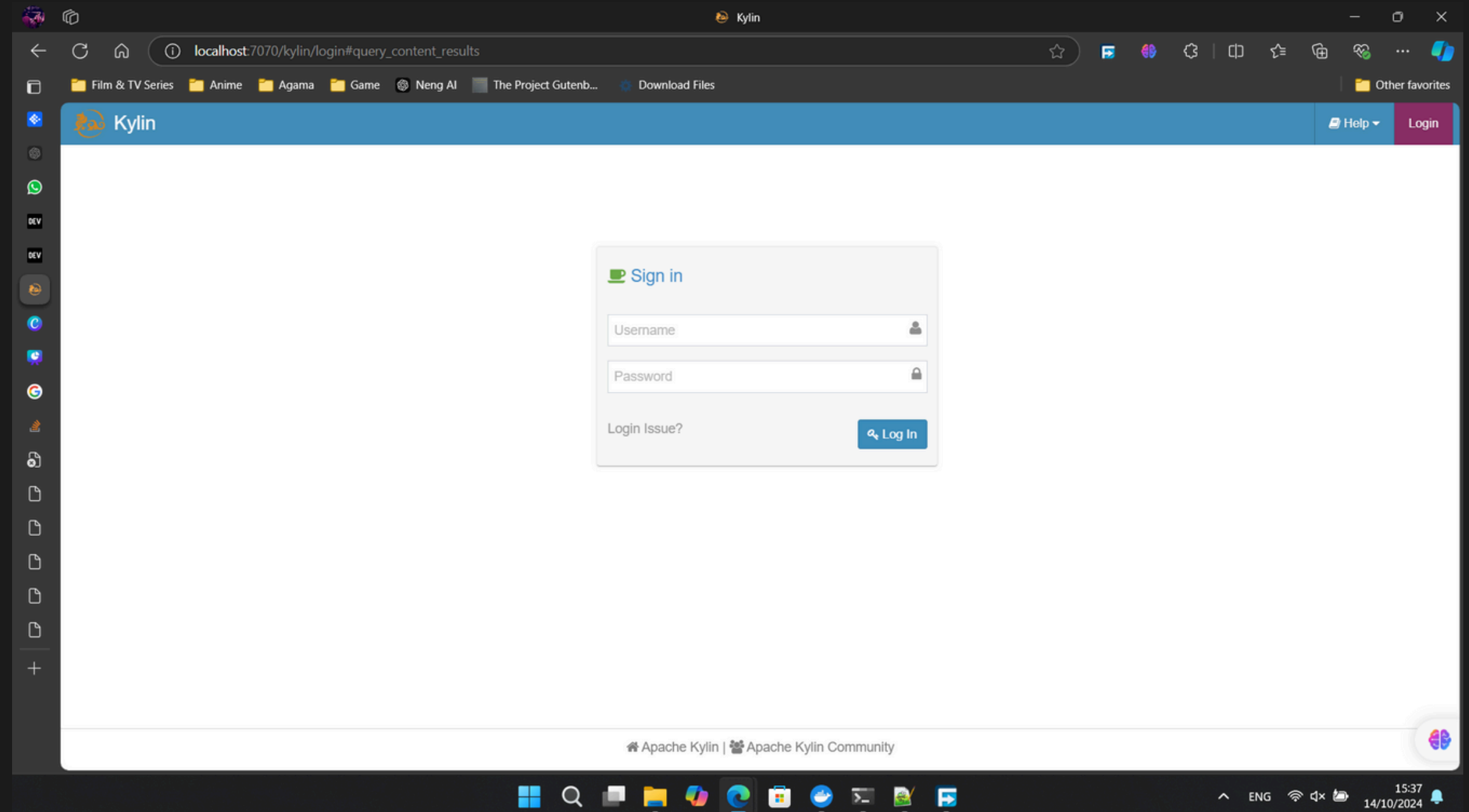


To open Kaylin you can write `http://localhost:7070/kylin/` in your URL

There is 3 Account you can use

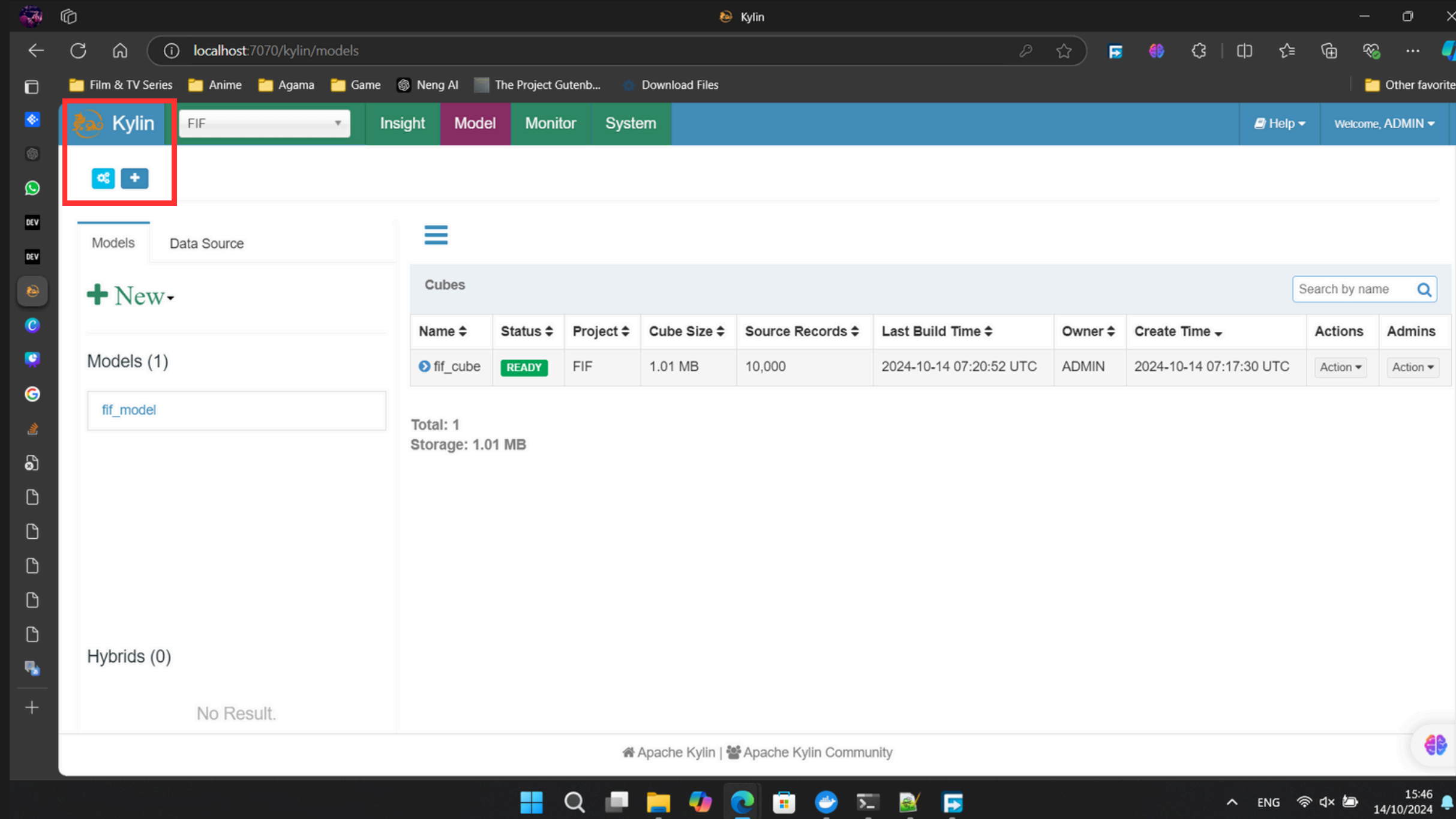
- Admin
Username : ADMIN
Password: KYLIN
- Modeler
Username: MODELER
Password: MODELER
- Analyst
Username: ANALYST
Password: ANALYST

Each account has its own
authority



Create A Project

After logging in to Kylin, click the + in the upper left corner to create a Project.



The project name is mandatory, while the project description is optional. However, a well-crafted project description can significantly facilitate future maintenance efforts.

New Project

Project Name

You can use letters, numbers, and underscore characters ' _ '

Project Description

Project Description..

Project Config

+ Property

Close

Submit

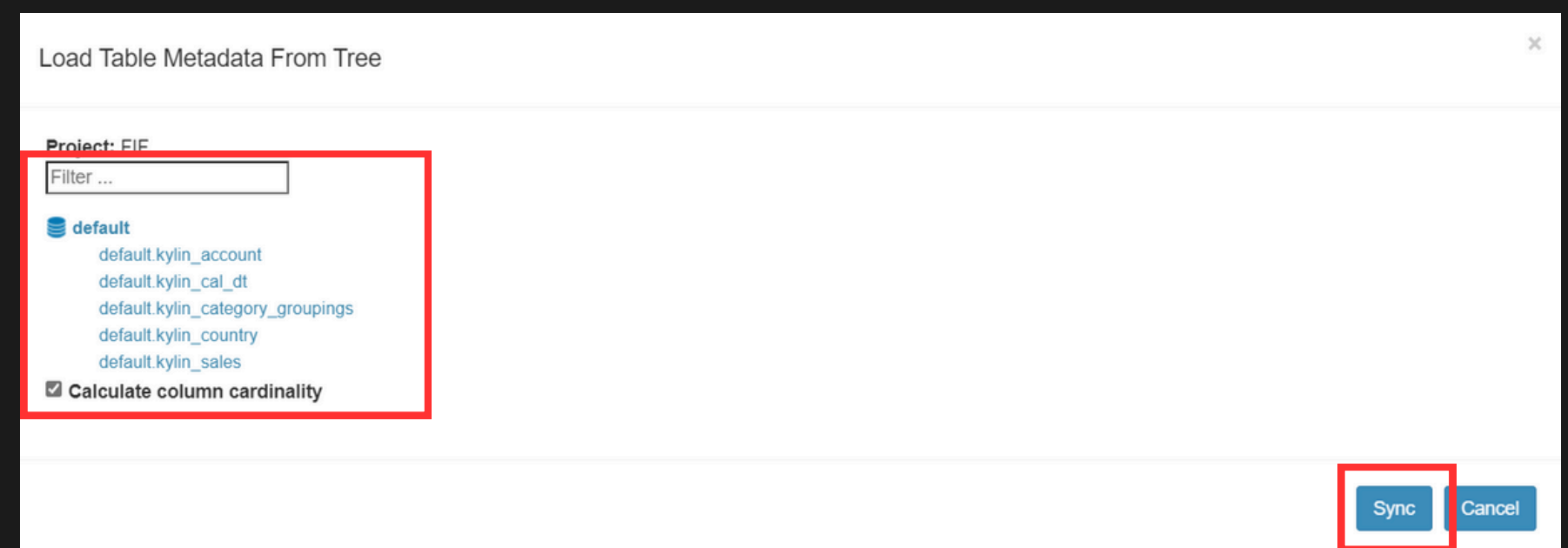
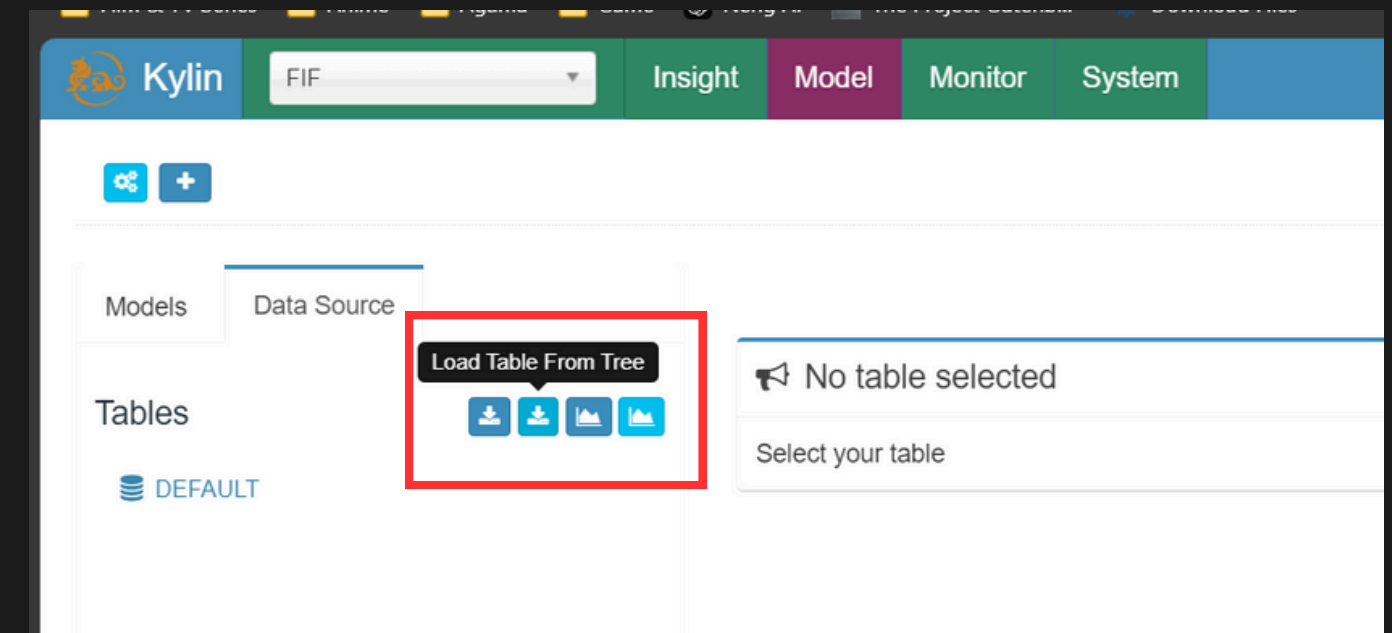
Add Data Source

To add a data source table to your project, follow these steps:

- Click the Add data source button in the upper left of the Data Asset -> Data Source interface.
- Select Hive as the data source type.
- Expand the database list to choose the desired data source.
- Click Sync.

Table sampling can help answer questions such as:

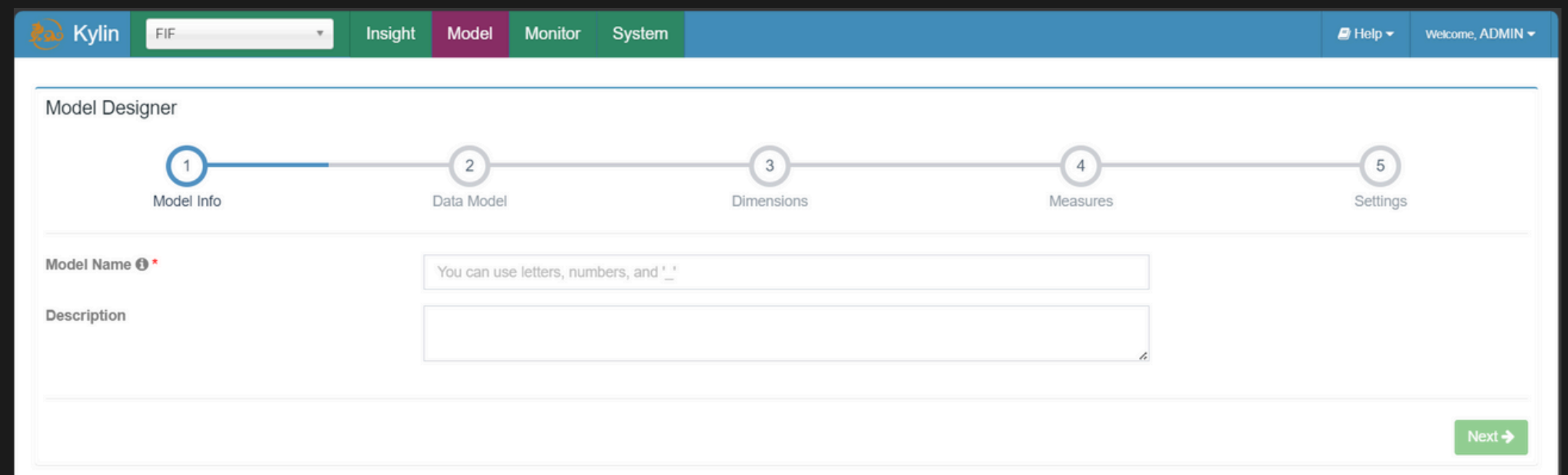
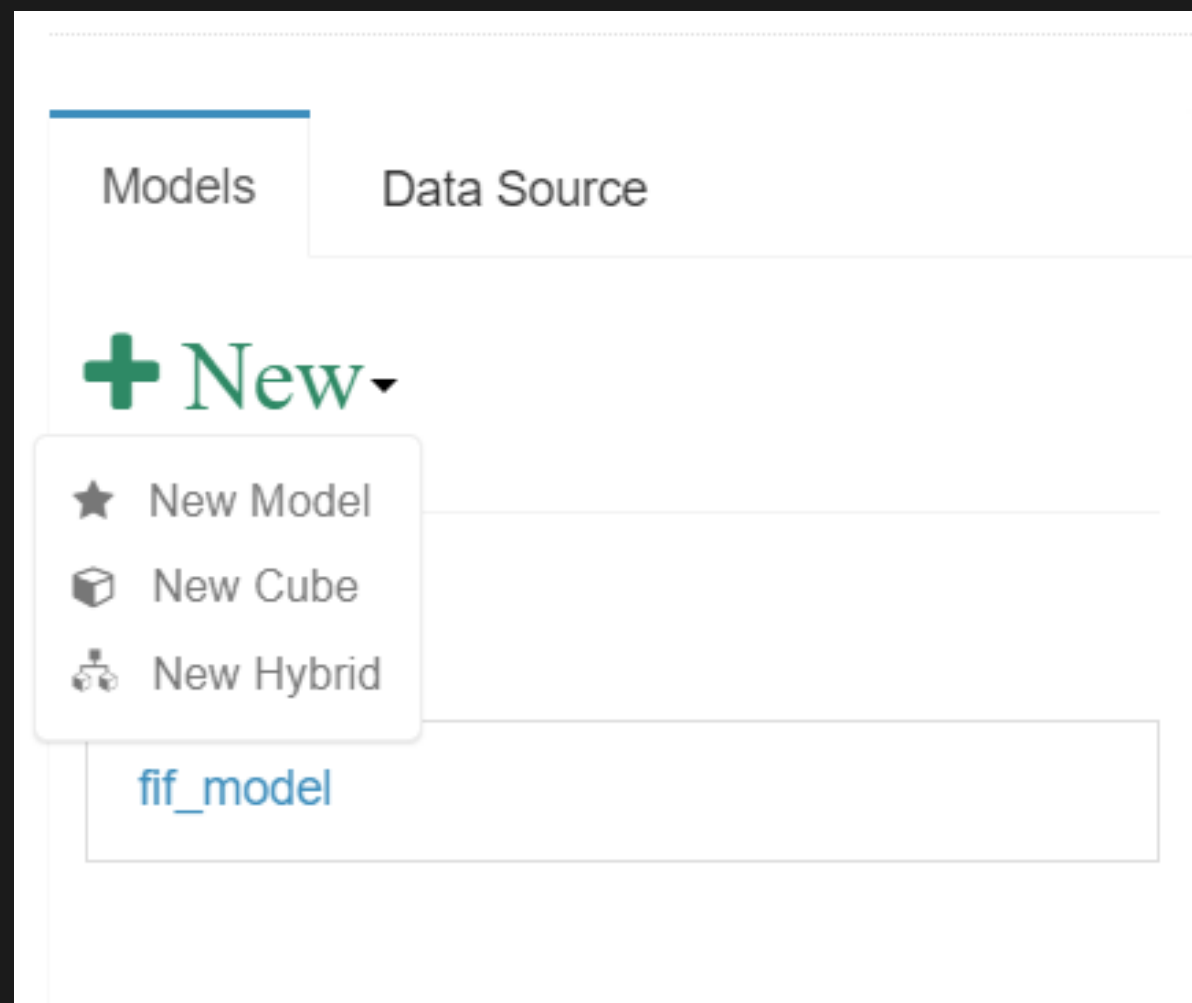
- Total number of rows in the table
- Cardinality of each column (amount of unique data)
- Characteristics of column values.



Create A Model

Model design is crucial in Kylin, significantly enhancing data analysis. It serves as the semantic layer, aiding users in visualizing business relationships among source tables.

To create a model you can click New -> New Model



Enter the Model Name and click Next. For this project you can name it "fif_model"

Choose the Fact Table and the Lookup Table. When incorporating the Lookup Table, it's essential to establish the JOIN condition with the Fact Table.

The screenshot displays the Kylin Model Designer interface. The top navigation bar includes the Kylin logo, a dropdown menu set to 'FIF', and tabs for 'Insight', 'Model' (active), 'Monitor', and 'System'. A 'Help' button and a user greeting 'Welcome, ADMIN' are also present. The 'Model Designer' section features a progress bar with five steps: 'Model Info' (completed), 'Data Model' (current), 'Dimensions', 'Measures', and 'Settings'. Below the progress bar, the 'Fact Table' section shows a '+ Add Lookup Table' button. A dropdown menu is open, listing available fact tables: 'DEFAULT.KYLIN_CAL_DT', 'DEFAULT.KYLIN_CATEGORY_GROUPINGS', 'DEFAULT.KYLIN_ACCOUNT', 'DEFAULT.KYLIN_COUNTRY', and 'DEFAULT.KYLIN_SALES'. To the right, the 'Add Lookup Table' dialog is shown. It includes a 'From Table' dropdown, a join type dropdown set to 'Inner Join', a 'Select Lookup Table' dropdown, and an 'AS' field for the 'Input Table Alias'. A checkbox for 'Skip snapshot for this lookup table' is also visible. A 'Tips' box on the right provides guidance: 1. Pick up a table joins another table that already exist. 2. Specify join relationship between two tables. 3. Join Type have to be same as will be used in query. The dialog has 'Cancel' and 'OK' buttons at the bottom right.

Fact Table: Typically, a fact table comprises quantifiable measures. A good example of a fact table is the order table, which includes columns such as order quantity and order amount that can be measured.

In this model we use the KYLIN_SALES

 Kylin

FIF

Insight

Model

Monitor

System

Help

Welcome, ADMIN

Model Designer

1

2

3

4

5

Model Info

Data Model

Dimensions

Measures

Settings

Fact Table *

DEFAULT.KYLIN_SALES

+ Add Lookup Table

Filter ...

ID	Table Alias	Table Name	Table Kind	Join Type	Join Condition	Actions
1	KYLIN_ACCOUNT	DEFAULT.KYLIN_ACCOUNT	Normal	inner	KYLIN_SALES.BUYER_ID = KYLIN_ACCOUNT.ACCOUNT_ID	
2	KYLIN_COUNTRY	DEFAULT.KYLIN_COUNTRY	Normal	inner	KYLIN_ACCOUNT.ACCOUNT_COUNTRY = KYLIN_COUNTRY.COUNTRY	

← Prev

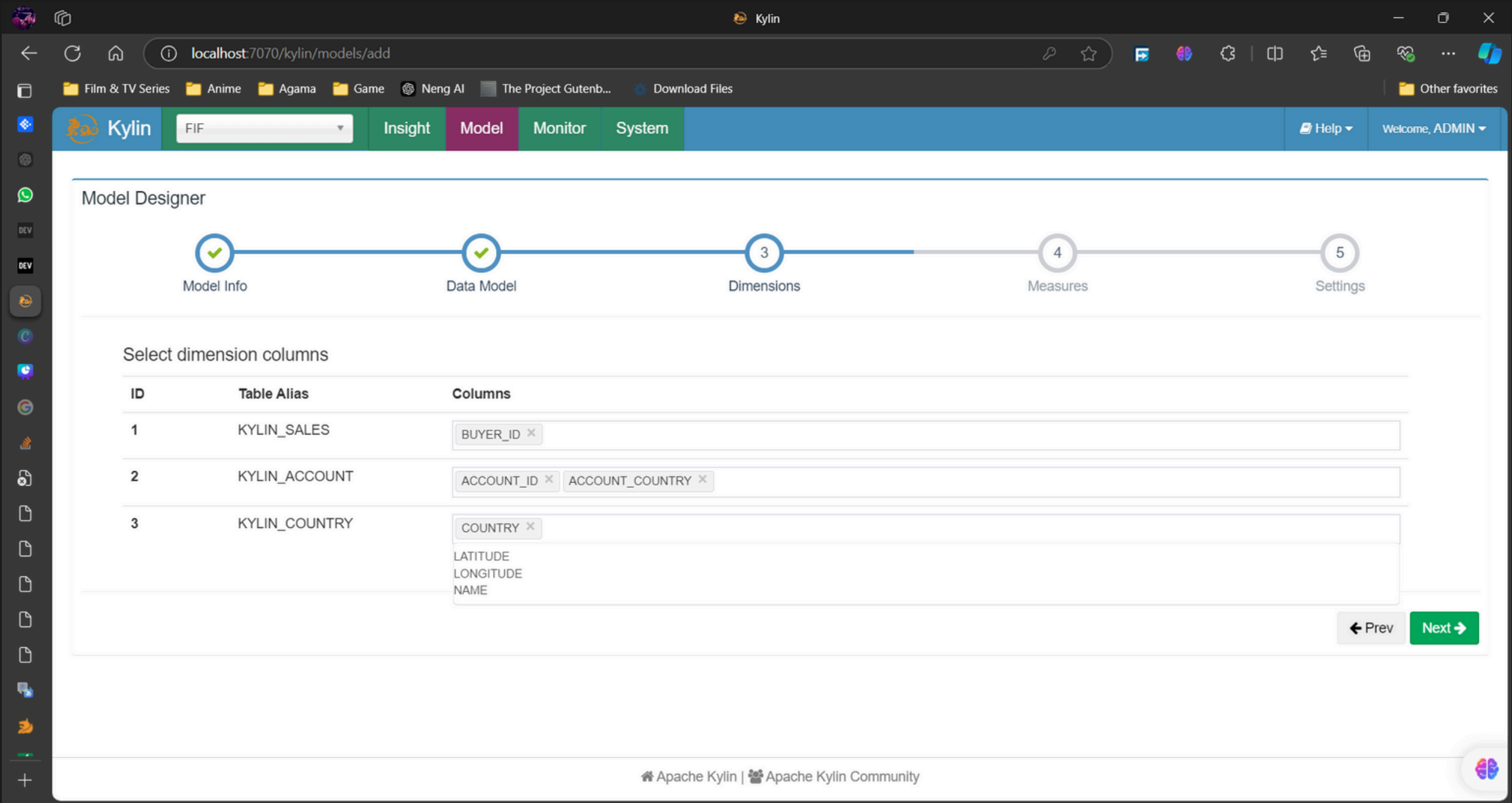
Next →

Apache Kylin

Apache Kylin Community

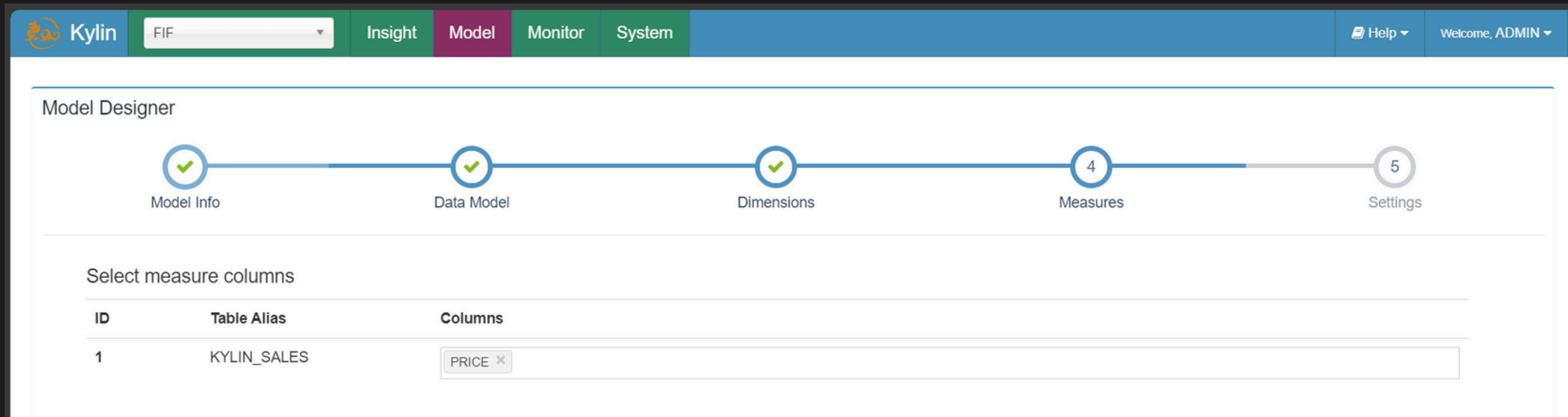


Then click Next to select Dimension:



Dimension Table: A table that represents a perspective looking into the quantifiable measures.

Next, Select Measure:



The screenshot shows the Kylin Model Designer interface. At the top, there is a navigation bar with tabs: Insight, Model (selected), Monitor, and System. To the right of the tabs are links for Help and a welcome message for ADMIN. Below the navigation bar, the 'Model Designer' section displays a workflow progress bar with five steps: Model Info, Data Model, Dimensions, Measures (current step, marked with a '4'), and Settings (marked with a '5'). The 'Measures' step is active, and the interface shows a table for selecting measure columns. The table has three columns: ID, Table Alias, and Columns. The first row shows ID '1', Table Alias 'KYLIN_SALES', and a selected column 'PRICE' with a close button (x).

ID	Table Alias	Columns
1	KYLIN_SALES	PRICE x

Measure table is a special table used to store and manage the calculation steps or measures applied to data in the model.

For this project we focus to try calculate the total price so we pick price

SETTINGS

- To set up time partitioning and filter conditions, define the time partition column to select the time range for incremental construction.
- If no column is set, all model cubes will be built. The filter condition serves as the WHERE clause when flattening the table.

The screenshot shows the Apache Kylin web interface at the URL `localhost:7070/kylin/models/add`. The interface has a top navigation bar with tabs: **Kylin**, **FIF**, **Insight**, **Model** (active), **Monitor**, and **System**. A breadcrumb trail at the top shows the progress: **Model Info** (checked), **Data Model** (checked), **Dimensions** (checked), **Measures** (checked), and **Settings** (5). The main content area is titled **Partition** and contains the following fields:

- Partition Date Column**: A dropdown menu showing `KYLIN_SALES` and `PART_DT`.
- Date Format**: A dropdown menu showing `yyyy-MM-dd`.
- Has a separate "time of the day" column ?**: A button labeled `No`.

Below the partition settings is a **Filter** section with a text area for the **WHERE** clause. The text area contains the placeholder text: "Please input WHERE clause without typing 'WHERE'".

At the bottom of the page, there is a footer with the text: `Apache Kylin | Apache Kylin Community`.

Then, click Save to save the model.

Create A Cube

Model -> New -> New Cube

Model Name : -> Pick Model you want in this one is fif_model

Cube Name : fif_cube

 Kylin

FIF

Insight

Model

Monitor

System

Help

Welcome, ADMIN

Models (1)

fif_model

Hybrids (0)

No Result.

1

Cube Info

2

Dimensions

3

Measures

4

Refresh Setting

5

Advanced Setting

6

Configuration Overwrites

7

Overview

Model Name *

fif_model

Cube Name *

fif_cube

Notification Email List

Comma Separated

Notification Events

ERROR x DISCARDED x SUCCEED x

Description

Next

Add Dimensions

To add a Dimension, click Next. Dimensions can be set as Normal or Derived, with the default being Derived. This means the column comes from the primary key of the dimension table, and only the primary key column will be calculated by the cube.

Auto Generate Dimensions ⓘ This is a helper for you to batch generate dimensions.
Visit [derived column](#) for more about derived column.

KYLIN_SALES [FactTable]

☒ Select All	Name	Columns
☒	BUYER_ID	BUYER_ID

KYLIN_ACCOUNT [LookupTable]

☒ Select All	Name	Columns		
☒	ACCOUNT_ID	ACCOUNT_ID	☐ Normal	☒ Derived
☒	ACCOUNT_COUNTRY	ACCOUNT_COUNTRY	☐ Normal	☒ Derived

KYLIN_COUNTRY [LookupTable]

☒ Select All	Name	Columns
--	------	---------

Cancel

OK

Add Measure

To add a pre-calculated measure, click Next and then + Measure. Kylin defaults to creating a Count (1) metric and supports eight metrics: SUM, MIN, MAX, COUNT, COUNT_DISTINCT, TOP_N, EXTENDED_COLUMN, and PERCENTILE.

Bulk Add Measures

Name measure name..	Expression	Parameters	Return Type	Actions
COUNT	COUNT	Value:1, Type:constant	bigint	 

Edit Measure

Name

TOTAL_PRICE

Expression ⓘ

SUM

Param Type

column

Param Value ⓘ

KYLIN_SALES.PRICE

☐ Also Show Dimensions

Return Type

DECIMAL(19,4)

OK

Cancel

Refresh Setting

After adding measures, click Next to access cube data refresh settings. Here, you can set the auto merge threshold, minimum data retention time, and the start time for the first segment.

Cube Designer

Progress: 1. Cube Info (✓) 2. Dimensions (✓) 3. Measures (✓) 4. Refresh Setting (4) 5. Advanced Setting (5) 6. Configuration Overwrites (6) 7. Overview (7)

Auto Merge Thresholds ⓘ

7	days	-
28	days	-

[New Thresholds +](#)

Volatile Range ⓘ

0

Retention Threshold ⓘ

0

Partition Start Date



Advance Setting

Advanced Settings, where you can configure the aggregation group, RowKeys, Mandatory Cuboids, and Cube Engine.

- For further details, refer to tutorial at http://kylin.apache.org/docs/tutorial/create_cube.html.
- For more dimensional optimization, visit: <http://kylin.apache.org/blog/2016/02/18/new-aggregation-group/>.

Cube Designer

Progress: 1. Cube Info (✓) 2. Dimensions (✓) 3. Measures (✓) 4. Refresh Setting (✓) 5. Advanced Setting (5) 6. Configuration Overwrites (6) 7. Overview (7)

Aggregation Groups

Visit [aggregation group](#) for more about aggregation group.

ID	Aggregation Groups	Max Dimension Combination: 0
1	Includes KYLIN_SALES.BUYER_ID x KYLIN_ACCOUNT.ACCOUNT_COUNTRY x Mandatory Dimensions Select Column... Hierarchy Dimensions New Hierarchy+ Joint Dimensions New Joint+	-

If you are not familiar with Advanced Settings, you can keep the default settings first.

Configuration Overwrites

For configuration items, please refer to: <http://kylin.apache.org/docs/install/configuration.html>.

Cube Designer

✓

Cube Info

✓

Dimensions

✓

Measures

✓

Refresh Setting

✓

Advanced Setting

6

Configuration Overwrites

7

Overview

Kylin Properties

+ Property

Tips

1. Cube level properties will overwrite configuration in kylin.properties

← Prev

Next →

After the configuration is complete, you can click the Save button to complete the cube creation.

Build Cube and Task Monitoring

The previously created cube has definitions but lacks calculated data and is "DISABLED." To enable data, you must build the cube by setting the start and end time, then clicking Submit. Monitor the build task status on the Monitor page. Once built, the cube's status will change to READY, allowing you to view segment information.

The screenshot displays the Kylin Monitor interface. At the top, the navigation bar includes 'Kylin', a dropdown menu set to 'FIF', and tabs for 'Insight', 'Model', 'Monitor' (active), and 'System'. On the right, there are links for 'Help' and 'Welcome, ADMIN'. Below the navigation bar, there are icons for settings and a plus sign. The main content area has two tabs: 'Jobs' (selected) and 'Slow Queries'. Under the 'Jobs' tab, there is a search bar for 'Cube Name' and a filter section for 'Jobs in' with options: 'LAST ONE DAY', 'ALL', 'NEW(0)', 'PENDING(0)', 'RUNNING(0)', 'STOPPED(0)', 'FINISHED(1)', 'ERROR(0)', and 'DISCARDED(0)'. A table lists the jobs, with one job highlighted: 'BUILD CUBE - fif_cube - 20120101000000_20211231235500 - UTC 2024-10-14 07:17:55'. The table columns are 'Job Name', 'Cube', 'Progress' (100%), 'Last Modified Time' (2024-10-14 07:20:58 UTC), 'Duration' (2.58 mins), and 'Actions'. To the right of the table is a 'Detail Information' panel showing job details: 'Job Name', 'Job ID', 'Project Name' (FIF), 'Status' (FINISHED), 'Built By', 'Duration' (2.58 mins), and 'MapReduce Waiting' (0.25 mins). At the bottom right, a 'Start' button shows the time '2024-10-14 07:18:23 UTC'.

Job Name	Cube	Progress	Last Modified Time	Duration	Actions
BUILD CUBE - fif_cube - 20120101000000_20211231235500 - UTC 2024-10-14 07:17:55	fif_cube	100%	2024-10-14 07:20:58 UTC	2.58 mins	Action

Total: 1

Start 2024-10-14 07:18:23 UTC

Insights and Query

In this section you can start query use MySQL to find anything you want

```
SELECT KYLIN_ACCOUNT.ACCOUNT_COUNTRY, SUM(KYLIN_SALES.PRICE) AS TOTAL_PRICE
FROM KYLIN_SALES
JOIN KYLIN_ACCOUNT
ON KYLIN_ACCOUNT.ACCOUNT_ID = KYLIN_SALES.BUYER_ID
GROUP BY KYLIN_ACCOUNT.ACCOUNT_COUNTRY
```

Like this one to find Total_Price every country

The results for its a Table but you can customise the visualization

Results (7)

	ACCOUNT_C..Y	TOTAL_PRICE...	
	DE	72034.5262	
	JP	70119.6708	
	CN	67048.9349	
	GB	74052.1557	
	IT	69723.7780	
	FR	69939.0301	
	US	72807.8659	

The screenshot shows the Kylin web interface. At the top, there's a navigation bar with tabs: Kylin, FIF, Insight, Model, Monitor, System, Help, and Welcome, ADMIN. Below the navigation bar, there's a sidebar with 'Tables' and 'DEFAULT'. The main area has tabs for 'New Query', 'Saved Queries', and 'Query History'. The 'New Query' tab is active, showing a query editor with a text area and a 'Submit' button. Below the query editor, there's a 'Results' section that says 'No Query Result.'.



Thank you...